

Review: Machines that think by Inga Strumke

Ben Herbst

I believe that basic knowledge about artificial intelligence will quickly become crucial for understanding much of what is happening in our lives and for being able to actively participate in the public debate about how we want this technology to influence our society.

— Inga Strumke

One has to be completely ignorant to be unaware of the debate raging about AI and how it is about to transform our societies. It doesn't help that the big-tech leaders and researchers have wide-ranging opinions, even among themselves. Here is a random selection:

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Artificial intelligence can be wondrous, but the technology underneath is more than a little monstrous. It eats up all the words in the world, from blogs to books, often without permission. It burns whole forests' worth of energy, digesting that raw material into its models, and gulps billions of gallons of water to cool down.

— Paul Ford

the Anthropic chief “knows absolutely nothing about the effects of technological revolutions on the labour market.

Yann LeCun, former chief scientist at Meta Platforms's AI division, has criticised Dario Amodei for warning that artificial intelligence (AI) could eliminate up to half of all tech jobs.

Responding to Amodei's earlier remarks, LeCun said the Anthropic chief “knows absolutely nothing about the effects of technological revolutions on the labour market,” adding that even AI leaders are not qualified to predict employment outcomes.

— Yann LeCun

When AI is everywhere, what knowledge, skills, and habits will students need to get ahead?

When AI is everywhere, what knowledge, skills, and habits will students need to get ahead? How can graduates be thoughtful, effective, purposeful users of this technology, benefitting from its value rather than becoming its casualties?

— Ben Wildavsky

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My original ambition in AI is to build what we sometimes call artificial general intelligence, which is a system that can exhibit all the cognitive capabilities that humans can. Then we know we have a truly general system at that point, a kind of Turing machine. There's still that to build, and I think that would be one of the most consequential inventions humanity will ever invent. I guess that was my original goal when I got into AI, 30 plus years ago and that's still my goal now. We still have a lot of work to do to get there.

— Demis Hassabis

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I think there are two concerns on AI for me. One is, these are general purpose systems and, like any powerful general purpose technology, it can be used for good, which of course, things like medicine and climate and help us with some of humanity's greatest challenges, but it can also be repurposed by bad actors for harmful ends. The technology itself is kind of neutral. It depends on how we as scientists and humanity and society decide are going to use it and deploy it. I worry about the bad actors and how they may access the same technologies for harmful ends. So that's one thing. The second thing is, as the systems become more and more powerful themselves, more autonomous, more agent like, can we understand and keep control of those systems, make sure they have the right goals and values and how do we ensure that the guardrails around how those systems work are robust enough?

— Demis Hassabis

The United States is at a pivotal moment.

The United States is at a pivotal moment. Multi-billion-dollar federal AI investments, rapid commercial deployment, and university-wide AI initiatives are collectively setting the trajectory of the nation's scientific, economic, and security landscape for decades to come.

At the same time: AI systems are being deployed faster than our ability to understand, verify, certify, or govern them. Federal agencies are launching large-

scale AI programs with expectations for near-term impact, often without proportional investment in mathematical foundations needed to understand, evaluate and systematically improve AI systems. Universities are shifting hiring, curricula, and research infrastructure around AI, often without commensurate investment in applied mathematics, with lasting workforce consequences. Global industrial leadership is shifting from those with the most data to those with the mathematical insight to transform raw AI potential into reliable, efficient, scalable, and sovereign economic power.

— SIAM AI Task Force Report

- Is AI a villain?
- or a saint?

I believe you get the idea - there are many reasoned ideas out there, but even more opinions. As for myself, I have no interest whatsoever in opinions. They muddy the waters and distract from the serious issues. It reminds me of something Tim Minchin once said:

Be hard on your opinions. We must think critically and not just about the ideas of others, be hard on your beliefs, take them out onto the verandah and hit them with a cricket bat. Be intellectually rigorous, identify your biases, your prejudices, your privileges.

— Tim Minchin

Of course this applies to me as well.

But, if the opinions are from tech leaders, they are consequential. Some of these leaders are incredibly successful and highly influential. Their opinions are consequential. What should our response be as members of societies with value systems that we hold dear? Doing nothing, risks being overwhelmed by a tsunami of sorts. And if we think a tsunami is not all good, how do we stop it? Or rather, direct it in a direction of our choosing where it can do the most good. Even if it is hard to direct a tsunami, it is important to realise that we ordinary people collectively have enormous power. Quoting Demis Hassabis again,

Human intelligence always astounds me, and I don't think we think about this enough. It's created modern civilisation around us. Sometimes when I'm flying over to the US for a business trip or something on a 747, I sometimes look out the window and think, "how have we as humanity manage this with our sort of primate brains?" It seems incredible to me, and I don't think people stop and think how magical that really is. So really the power of intelligence, human intelligence, has created the wonders of modern civilisation.

— Demis Hassabis

We must never surrender our intellectual autonomy.

The thing is, we as humans have amazing intelligence with almost unlimited potential. We might even be able to build machines smarter than ourselves! We must never surrender our intellectual autonomy. But the question remains, how do we respond to the AI revolution? One answer is to use our wondrous intellect to equip ourselves to participate in a wide, informed debate of where we want to go, and how to get there. *Machines that Think* is an attempt to provide some of that equipment.

With MtT Inga managed to write a book that is both informative and entertaining. She writes in a lively, engaging style and her passion for AI shines through the whole book. Most of her explanations of technical issues are brilliant. Yes, she does not shy away from technicalities when it is necessary to follow the broader argument. One exception is her discussion of Gödel's incompleteness theorem (Nagel & Newman, 1958) which I found pretty hard going. For the intrepid who wants to know more, Natalie Wolchover provides [an accessible overview](#), or you may want to go directly to her source, Nagel and Newman.

REFERENCES

Nagel, E., & Newman, J. R. (1958). *Gödel's Proof*. New York University Press.